

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						***** VALVES *****			817	USE ONLY FOR TIE IN ALL BOLTS ARE METRIC DIA. 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
20	U04						***** MODULAR VALVES *****				
30											
30	U04		MODULAR DBB	M06AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
40	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
50										SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
50	U04						***** BALL VALVES *****				
60											
60	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
70	U04		BALL	B06AC05	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0		
80	U04		BALL	B06AC07	4	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
90											
90	U04						***** RISING STEM BALL VALVES *****				
100											
100	U04	VA5	BALL	R06AC06	2	6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
110	U04	VA5	BALL	R06AC08	8	10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
120											
120	U04						***** GATE VALVES *****				
130											
130	U04		GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
140	U04		GATE	W06AC71	6	24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
150	U04	817	GATE	W06CC71	28	28	GATE 600# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
160											
160	U04						***** PIPE ABOVE GROUND & BELOW GROUND*****				
170											
170	U04		PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6.PE	0	0		
180	U04		PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
190	U04		PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
200											
200	U04		PIPE	9001208CTD	4	4	LINEPIPE SMLS WT= 5.16MM API-5L X60BE COATED	0	0		
210	U04		PIPE	9001254CTD	6	8	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
220	U04		PIPE	9001496CTD	10	10	LINEPIPE SMLS WT = 7.80 MM API-5L X60 BE COATED	0	0		
230	U04		PIPE	9001498CTD	12	12	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE COATED	0	0		
240	U04		PIPE	9001487CTD	14	14	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
250	U04		PIPE	9001493CTD	16	16	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
260	U04		PIPE	9001494CTD	18	18	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED	0	0		
270	U04		PIPE	9001488CTD	20	20	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED	0	0		
280	U04		PIPE	9001495CTD	24	24	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED	0	0		
290	U04		PIPE	9001329CTD	28	28	LINEPIPE DSAW WT= 14.27 MM API-5L X60 BE COATED	0	0		
300	U04		PIPE	9001498	12	12	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE	0	0		
310											
310	U04						*****FITTINGS ABOVE & BELOW GROUND GROUND*****				
320											
320	U04		ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
330	U04		ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
340	U04		ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
350	U04		ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
360											
360	U04		TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
370	U04		TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
380	U04		TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2		
390	U04		TEE RED	5392454	0.75	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
400											
400	U04		CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
410	U04		CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
420	U04		CAP	9005074	4	4	CAP BW WT 5.16 MM LTCS A860 WPHY-60	0	0		
430	U04		CAP	9005055	6	8	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0		
440	U04		CAP	9005075	10	10	CAP BW WT 7.80 MM LTCS A860 WPHY-60	0	0		
450	U04		CAP	9005076	12	12	CAP BW WT 8.38 MM LTCS A860 WPHY-60	0	0		
460	U04		CAP	9005059	14	14	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
470	U04		CAP	9005060	16	16	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0		
480	U04		CAP	9005061	18	18	CAP BW WT 10.31 MM LTCS A860 WPHY-60	0	0		
490	U04		CAP	9005064	20	20	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0		
500	U04		CAP	9005065	22	22	CAP BW WT 11.91 MM LTCS A860 WPHY-60	0	0		
510	U04		CAP	9005067	24	24	CAP BW WT 12.70 MM LTCS A860 WPHY-60	0	0		
520	U04		CAP	9005068	26	28	CAP BW WT 14.27 MM LTCS A860 WPHY-60	0	0		
530											
530	U04		RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2		
540	U04		RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2		
550	U04		RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	600K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
560	U04		RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3	
570	U04		RED ECC	9005077	3	3	REDUCER ECC WT 5.16 MM / SCH STD A860 WPHY 60	4	4	
580	U04		RED ECC	9006103	3	3	REDUCER ECC WT 6.35 MM / STD LTCS A860 WPHY-60	6	6	
590	U04		RED ECC	9002385	4	4	REDUCER ECC WT 6.35 MM / 5.16 MM LTCS A860 WPHY-60	6	8	
600	U04		RED ECC	9002266	6	6	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	8	
610	U04		RED ECC	9002266	10	10	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	12	12	
620	U04		RED ECC	9002386	4	4	REDUCER ECC WT 7.80 MM / 5.16 MM LTCS A860 WPHY-60	10	10	
630	U04		RED ECC	9002387	6	8	REDUCER ECC WT 7.80 MM / 6.35 MM LTCS A860 WPHY-60	10	10	
640	U04		RED ECC	9002388	6	8	REDUCER ECC WT 8.38 MM / 6.35 MM LTCS A860 WPHY-60	12	12	
650	U04		RED ECC	9002390	6	8	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	14	14	
660	U04		RED ECC	9002391	10	10	REDUCER ECC WT 8.74 MM / 7.80 MM LTCS A860 WPHY-60	14	14	
670	U04		RED ECC	9002392	12	12	REDUCER ECC WT 8.74 MM / 8.38 MM LTCS A860 WPHY-60	14	14	
680	U04		RED ECC	9002393	8	8	REDUCER ECC WT 9.53 MM / 6.35 MM LTCS A860 WPHY-60	16	16	
690	U04		RED ECC	9002394	10	10	REDUCER ECC WT 9.53 MM / 7.80 MM LTCS A860 WPHY-60	16	16	
700	U04		RED ECC	9002395	12	12	REDUCER ECC WT 9.53 MM / 8.38 MM LTCS A860 WPHY-60	16	16	
710	U04		RED ECC	9002396	14	14	REDUCER ECC WT 9.53 MM / 8.74 MM LTCS A860 WPHY-60	16	16	
720	U04		RED ECC	9002397	10	10	REDUCER ECC WT 10.31 MM / 7.80 MM LTCS A860 WPHY-60	18	18	
730	U04		RED ECC	9002398	12	12	REDUCER ECC WT 10.31 MM / 8.38 MM LTCS A860 WPHY-60	18	18	
740	U04		RED ECC	9002399	14	14	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60	18	18	
750	U04		RED ECC	9002400	16	16	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60	18	18	
760	U04		RED ECC	9004160	12	12	REDUCER ECC WT 11.13 MM / 8.38 MM LTCS A860 WPHY-60-W	20	20	
770	U04		RED ECC	9004161	14	14	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60-W	20	20	
780	U04		RED ECC	9004162	16	16	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60-W	20	20	
790	U04		RED ECC	9004163	18	18	REDUCER ECC WT 11.13 MM / 10.31 MM LTCS A860 WPHY-60-W	20	20	
800	U04		RED ECC	9004164	16	16	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60-W	24	24	
810	U04		RED ECC	9004165	18	18	REDUCER ECC WT 12.70 MM / 10.31 MM LTCS A860 WPHY-60-W	24	24	
820	U04		RED ECC	9004166	20	20	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60-W	24	24	
830	U04		RED ECC	9002408	18	18	REDUCER ECC WT 14.27 MM / 10.31 MM LTCS A860 WPHY-60-W	28	28	
840	U04		RED ECC	9004167	20	20	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60-W	28	28	
850	U04		RED ECC	9004168	24	24	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60-W	28	28	
860										
860	U04						***** INDUCTION BENDS ABOVE GROUND*****			
870										
870	U04		INDUC BEND	9001819	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
880	U04		INDUC BEND	9001813	6	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
890	U04		INDUC BEND	9001829	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
900	U04		INDUC BEND	9001839	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
910	U04		INDUC BEND	9001816	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
920	U04		INDUC BEND	9001817	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
930	U04		INDUC BEND	9001818	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
940	U04		INDUC BEND	9001899	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
950	U04		INDUC BEND	9005321	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
960	U04		INDUC BEND	9005322	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
970	U04		INDUC BEND	9005323	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
980	U04		INDUC BEND	9001823	6	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
990	U04		INDUC BEND	9005324	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1000	U04		INDUC BEND	9005325	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1010	U04		INDUC BEND	9001826	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1020	U04		INDUC BEND	9001827	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1030	U04		INDUC BEND	9001828	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1040	U04		INDUC BEND	9005326	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1050	U04		INDUC BEND	9005327	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1060	U04		INDUC BEND	9005328	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1070	U04		INDUC BEND	9005329	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1080	U04		INDUC BEND	9001833	6	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1090	U04		INDUC BEND	9005330	10	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1100	U04		INDUC BEND	9003260	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1110	U04		INDUC BEND	9001836	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1120	U04		INDUC BEND	9001836	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1130	U04		INDUC BEND	9001837	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1140	U04		INDUC BEND	9001838	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1150	U04		INDUC BEND	9005333	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1160	U04		INDUC BEND	9005334	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1170										
1170	U04						***** INDUCTION BENDS BELOW GROUND*****			
1180										
1180	U04		INDUC BEND	9001819CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1190	U04		INDUC BEND	9001813CTD	6	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1200	U04		INDUC BEND	9001829CTD	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1210	U04		INDUC BEND	9001839CTD	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1220	U04		INDUC BEND	9001816CTD	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1230	U04		INDUC BEND	9001817CTD	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: - / 40 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	600K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
1240	U04		INDUC BEND	9001818CTD	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1250	U04		INDUC BEND	9001899CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1260	U04		INDUC BEND	9005321CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1270	U04		INDUC BEND	9005322CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1280	U04		INDUC BEND	9005323CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1290	U04		INDUC BEND	9001823CTD	6	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1300	U04		INDUC BEND	9005324CTD	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1310	U04		INDUC BEND	9005325CTD	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1320	U04		INDUC BEND	9001826CTD	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1330	U04		INDUC BEND	9001827CTD	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1340	U04		INDUC BEND	9001828CTD	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1350	U04		INDUC BEND	9005326CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1360	U04		INDUC BEND	9005327CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1370	U04		INDUC BEND	9005328CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1380	U04		INDUC BEND	9005329CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1390	U04		INDUC BEND	9001833CTD	6	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1400	U04		INDUC BEND	9003260CTD	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1410	U04		INDUC BEND	9001836CTD	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1420	U04		INDUC BEND	9001836CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1430	U04		INDUC BEND	9001837CTD	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1440	U04		INDUC BEND	9001838CTD	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1450	U04		INDUC BEND	9005334CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1460														
1460	U04						*****OLETS ABOVE & BELOW GROUND*****							
1470														
1470	U04		OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	8					
1480	U04		OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8	8					
1490	U04		OLETS	9002452	0.75	2	WELDOLET WT 7.80 MM/SCH XS WT LTCS A694 F60	10	10					
1500	U04		OLETS	9005084	3	3	WELDOLET WT 7.80 MM/STD WT LTCS A694 F60	10	10					
1510	U04		OLETS	9002453	0.75	2	WELDOLET WT 8.38 MM/SCH XS WT LTCS A694 F60	12	12					
1520	U04		OLETS	9005085	3	3	WELDOLET WT 8.38 MM/STD WT LTCS A694 F60	12	12					
1530	U04		OLETS	9006423	0.75	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	14	14					
1540	U04		OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	14	14					
1550	U04		OLETS	9002456	4	4	WELDOLET WT 8.74MM/ 5.16 MM LTCS A694 F60	14	14					
1560	U04		OLETS	9006429	0.75	2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	16	16					
1570	U04		OLETS	9006428	3	3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	16	16					
1580	U04		OLETS	9002458	4	4	WELDOLET WT 9.53 MM/5.16 MM WT LTCS A694 F60	16	16					
1590	U04		OLETS	9002459	0.75	2	WELDOLET WT 10.31 MM/ SCH XS WT LTCS A694 F60	18	18					
1600	U04		OLETS	9006434	3	3	WELDOLET WT 10.31 MM/STD WT LTCS A694 F60	18	18					
1610	U04		OLETS	9002461	4	4	WELDOLET WT 10.31 MM/ 5.16 WT LTCS A694 F60	18	18					
1620	U04		OLETS	9005119	0.75	2	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	20	20					
1630	U04		OLETS	9005086	3	3	WELDOLET WT 11.13 MM/STD WT LTCS A694 F60	20	20					
1640	U04		OLETS	9005089	4	4	WELDOLET WT 11.13 MM/ 5.16 MM WT LTCS A694 F60	20	20					
1650	U04		OLETS	9005120	6	6	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	20	20					
1660	U04		OLETS	9002467	0.75	2	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	24	24					
1670	U04		OLETS	9005087	3	3	WELDOLET WT 12.70 MM/STD WT LTCS A694 F60	24	24					
1680	U04		OLETS	9005087	3	3	WELDOLET WT 12.70 MM/STD WT LTCS A694 F60	28	28					
1690	U04		OLETS	9002468	4	4	WELDOLET WT 12.70 MM/ 5.16 MM WT LTCS A694 F60	24	24					
1700	U04		OLETS	9002469	6	8	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	24	24					
1710	U04		OLETS	9002471	0.75	2	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	28	28					
1720	U04		OLETS	9002472	4	4	WELDOLET WT 14.27 MM/ 5.16 MM LTCS A694 F60	28	28					
1730	U04		OLETS	9002473	6	8	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	28	28					
1740	U04		OLETS	9002474	10	10	WELDOLET WT 14.27 MM/ 7.80 MM LTCS A694 F60	28	28					
1750														
1750	U04		OLETS	5504207	0.75	2	NIPOFLANGE 600# RTJ SCH STD/XS LTCS A350 LF2 CL.1	3	3					
1760	U04		OLETS	9005079	0.75	2	NIPOFLANGE 600# RF 5.16 MM/XS WT LTCS A694 F60	4	4					

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	600K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1770	U04		OLETS	9004270	0.75	2	NIPOFLANGE 600# RF 6.35 MM/SCH XS WT LTCS A694 F60	6	8	
1780	U04		OLETS	9005080	0.75	2	NIPOFLANGE 600# RF 7.80 MM/XS WT LTCS A694 F60	10	10	
1790	U04		OLETS	9005081	0.75	2	NIPOFLANGE 600# RF 8.38 MM/XS WT LTCS A694 F60	12	12	
1800	U04		OLETS	9005032	0.75	2	NIPOFLANGE 600# RF 8.74 MM/XS WT LTCS A694 F60	14	14	
1810	U04		OLETS	9005082	0.75	2	NIPOFLANGE 600# RF 9.53 MM/XS WT LTCS A694 F60	16	16	
1820	U04		OLETS	9005033	0.75	2	NIPOFLANGE 600# RF 10.31 MM/XS WT LTCS A694 F60	18	18	
1830	U04		OLETS	9005102	0.75	2	NIPOFLANGE 600# RF 11.13 MM/XS WT LTCS A694 F60	20	20	
1840	U04		OLETS	9005104	0.75	2	NIPOFLANGE 600# RF 12.70 MM/XS WT LTCS A694 F60	24	24	
1850	U04		OLETS	9005105	0.75	2	NIPOFLANGE 600# RF 14.27 MM/XS WT LTCS A694 F60	28	28	
1860										
1860	U04						***** FLANGES ABOVE & BELOW GROUND*****			
1870										
1870	U04		FLANGE	5557036	0.75	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1880	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1890	U04		FLANGE	9003480	4	4	FLG WN 600# RF LTCS A694 F60 WT 5.16MM	0	0	
1900	U04		FLANGE	9001471	6	8	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0	
1910	U04		FLANGE	9003482	10	10	FLG WN 600# RF LTCS A694 F60 WT 7.80 MM	0	0	
1920	U04		FLANGE	9003483	12	12	FLG WN 600# RF LTCS A694 F60 WT 8.38 MM	0	0	
1930	U04		FLANGE	9003484	14	14	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0	
1940	U04		FLANGE	9003485	16	16	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM	0	0	
1950	U04		FLANGE	9003486	18	18	FLG WN 600# RF LTCS A694 F60 WT 10.31 MM	0	0	
1960	U04		FLANGE	9003487	20	20	FLG WN 600# RF LTCS A694 F60 WT 11.13 MM	0	0	
1970	U04		FLANGE	9001512	24	24	FLG WN 600# RF LTCS A694 F60 WT 12.70MM	0	0	
1980	U04	817	FLANGE	9001753	28	28	FLG COMPACT 600# RJ A694 F60 WT 14.27 MM	0	0	
1990	U04		BLIND	5615598	0.75	3	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
2000	U04		BLIND	9001518	4	24	FLG BLIND 600# RF LTCS A694 F60	0	0	
2010	U04		FLANGE	9003489	28	28	FLG WN 600# RF LTCS A694 F60 WT 14.27 MM SERIES A	0	0	
2020										
2020	U04						***** GASKETS*****			
2030										
2030	U04		GASKET	5673103	0.75	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IIR	0	0	
2040	U04	817	GASKET	5673300	28	28	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 600K5D	0	0	
2050	U04		GASKET	5674012	28	28	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2060										
2060	U04	B1					***** BOLTING*****			
2070										
2070	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2080										
2080	U04	SB6	BOLTING	5675675	10	28	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2090										
2090	U04						***** BLINDS & SPACERS*****			
2100										
2100	U04		SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
2110										
2110	U04		PADDLE	5750040	10	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
2120	U04		PADDLE	5750251	10	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
2130	U04		PADDLE	5750030	28	28	PADDLE BLANK 600# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
2140	U04		PADDLE	5750252	28	28	PADDLE SPACER 600# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
2150										

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	600K5D